

Gabriel Lucchesi

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EDUCATION

Florida International University, Honors College

Miami, FL

Bachelor of Science in Computer Science — GPA: 4.0/4.0

Aug. 2022 – December 2025

EXPERIENCE

Google - YouTube

May 2025 - August 2025

Software Engineering Intern

Mountain View, CA

- Researched applications of LLM post-training methods for comment and channel-wide **spam detection** at scale, overall improving by **53%** the number of caught live-traffic impressions in adversarial view rate (AVR) samples.
- Designed custom **reinforcement learning**-based reward models to enhance reasoning and OOD generalization for adversarial spam using the Group-Relative Policy Optimization (**GRPO**) algorithm and its latest SOTA variants.
- Created a **JAX**-based library to fine-tune Gemma models w/ **LoRA & Reasoning SFT** on classification tasks, supporting both **model and data sharding** strategies for higher throughput under low TPU HBM constraints.

NVIDIA

May 2024 - August 2024

Software Engineering Intern

Santa Clara, CA

- Drove development of key user-facing modules in OSMO, a cloud-based **ML infrastructure** platform for robotics & GPU scheduling, delivering the platform's first productized UI and greatly increasing daily active users (DAUs).
- Designed workflow-scheduling APIs in **Python** to be integrated with a **Bazel** and **Kubernetes** backend.
- Assisted **GEAR**'s research of general video-game agents by analyzing data for training inverse dynamics models.

INIT FIU (Volunteering)

March 2023 – Present

Director of Technology

Miami, FL

- Created the website orchestrating ShellHacks 2024, Florida's largest Hackathon, with over **1,400** student hackers.
- Utilized **Notion**'s API to create a **Judging System** in **Streamlit** for Sponsors and Organizers to use, reducing judging latency by **250%** through effective database operations and data visualization on project submissions.

RESEARCH

Defense Advanced Research Projects Agency

September 2024 - April 2025

Undergraduate Researcher (Contract)

Miami, FL

- Integrated early fusion-based models into robots in day-to-day embodied scenarios as **reinforcement learning** policies to complete reasoning-intensive tasks instructed via natural language and egocentric camera input.
- Researched efficient **sim-to-sim transfer learning** in **embodied AI models** from Habitat-Sim into Isaac Sim.
- Aligned multimodal LLMs' hidden states with representations of previous High Level policy actions in PDDL.

Florida International University

August 2023 – April 2024

Undergraduate Researcher (Advanced Research and Creativity in Honors)

Miami, FL

- Applied attribution algorithms for **Explainable AI** in agent-based systems of deep reinforcement learning under Sumit Kumar Jha, Ph.D. for improving reliability and transparency of autonomous control systems.
- Trained a lunar lander agent in a Box2D environment of **Farama Foundation's Gymnasium** using **Deep Deterministic Policy Gradients** for simultaneous control of engine thrusters and smoother landing.
- Leveraged **PyTorch**'s *Captum.ai* to implement **Integrated Gradients** algorithms upon action inference.
- Implemented a HuggingFace Space to visualize the attributions of the agent's decision to the input space.

SELECTED PUBLICATIONS

- [1] From Signals to Semantics: A Survey on Time Series Explainability through a Human-Cognitive Lens
Chen, Zhuomin, **Lucchesi, Gabriel**, Dong, Qingkai, Zheng, Xu, Song, Dongjin, Wen, Qingsong, Cheng, Wei, Ni, Jingchao, and Luo, Dongsheng, *Under review at the Journal of the ACM, 2025*

SELECTED ORAL PRESENTATIONS

National Conference for Undergraduate Research

April 2024 | (Proceedings)

Opening The Black Box: Explainability and Introspection in Deep Reinforcement Learning

TEACHING

- COP 3337 (Programming II), MAC 2311 (Calculus I):** Teaching Assistant Spring 2023 - Spring 2024
- Attended to over **120** students in total, hosting weekly office hours, helping in lectures and assisting in grading.
- AI Guild at FIU:** Coordinator and Lead Instructor | [Curriculum](#) August 2023 - Present
- Hosted voluntary workshops on **Deep Learning**, **CUDA**, Linear Algebra, **JAX**, and **PyTorch** for a community of **35+** students, collectively reading and implementing research papers as part of semi-weekly projects.
 - Spotlight Projects: [GPT-2 124M from scratch](#), [Variational Auto-Encoder](#), [Mini Tensor Autodiff module](#)

PROJECTS

- Attention Mechanisms** | *JAX, Flax, WandB* March 2025 | ([GitHub](#), [Pre-arXiv](#))
- Co-wrote and implemented “A Survey on State-of-the-art Attention Mechanisms” in **JAX** and **Flax** Linen API.
 - Engineered **neural attention modules** for training **transformer models** based on four architectures: Multi-Headed Self-Attention, Grouped Query Attention, Multi-Query Attention, & Multi-Head Latent Attention.
 - Applied **KV-Caching** optimizations for incremental attention at inference while maintaining tensor consistency.
- Aura** | *PyTorch, CLIP, Rust, Generative Adversarial Networks, CNNs* September 2024 - Present | ([GitHub](#))
- Lead **7 students** into training a live, light-weight **StyleGAN model** for conditional video art generation.
 - Created a face detection and emotion **CNN encoder** pipeline with **PyTorch**, reaching a **0.56s** inference time.
 - Aligned emotion-vector embeddings with synthetic video frames using **CLIP**-like alignment at a high Recall @ K.
 - Implemented **WebRTC** in **Rust** for camera I/O between the model and user, achieving **near real-time** speed.
- Memora** | *Python, LangChain, IrisDB, Llama 3.1b, Flask* September 2024 | ([GitHub](#))
- Leveraged **RAG** to co-develop a memory management system in IrisDB for Alzheimer’s patients, using facial recognition with **OpenCV**, and **image-to-text** processing with **Qwen2-VL**.
 - Used **LangChain** to fetch multimodal, relevant information to a Llama 3.1b personal assistant given a query.
 - Awarded **1st place** at HackMIT’s “Best Use of Vector Search for Generative AI” challenge.
- Mimir** | *Go, Python, RAPIDS cuML, WeaviateDB* September 2024 | ([GitHub](#))
- Collaborated to create a ServiceNow accelerator **recommendation engine** based on company reports and needs.
 - Implemented NLP-based **TF-IDF** and **K-Means Clustering** with **cuML** GPU support to categorize reports.
 - Leveraged **WeaviateDB** to create a chat system for accelerator-based **RAG** & chat history rendering in Next.js.
 - Awarded **1st place** for ServiceNow’s challenge at **University of Central Florida**’s hackathon, KnightHacks
- Project DARWIN** | *PyTorch, MuJoCo, Gymnasium* September 2023 – December 2023 | ([GitHub](#))
- Led a 6-person team to create a project aimed at exploring OpenAI Gyms, MuJoCo and **Deep RL methods**.
 - Successfully trained policy networks utilizing **Q-learning**, **Deep-Q Learning** and **DDPG** algorithms.
 - Used wrappers to create a universal framework that teaches agents to play 100+ Atari games from GUI input.
- Pots n’ Pans** | *Computer Vision, RoboFlow, Python, Next.js* October 2023 | ([GitHub](#))
- Created a **Next.js** application with **Computer Vision** to get recipes from ingredients and minimize food waste.
 - Deployed a **YOLOv8 CNN model** under a **serverless GPU architecture** in **RunPod** to identify ingredients in a picture, trained with **image segmentation** on fridge items to query a database for over **300,000+** recipes.
 - Awarded **1st place** for sustainability at **Georgia Institute of Technology**’s yearly hackathon, Hack GT.

TECHNICAL SKILLS

Programming Languages: Python, Java, Rust, Golang, C++, SQL, Swift, JavaScript, TypeScript
ML: JAX, PyTorch, TensorFlow, Gymnasium, MuJoCo, SK-Learn, Keras, CUDA, DSPy, LangChain, Pydantic
Developer Tools: Bazel, Jenkins, Docker, Kubernetes, Hugging Face, Git, GCP & Vertex AI, AWS & Azure
Data Science: Pandas, NumPy, Matplotlib, SciPy, Seaborn, Jupyter Notebooks
Languages Spoken: English (Fluent, C2), Brazilian Portuguese (Native), Spanish (Proficient), French (Proficient)

HONORS AND AWARDS

- Outstanding Engineering Graduate Award** Florida International University
Awarded to 1 student per graduating class recognizing academic excellence Fall 2025
- Presidential Merit Scholarship Award** Florida International University
Full cost of attendance covered by institution Fall 2022 - Fall 2025
- Dean’s List: Summa Cum Laude** Florida International University
Awarded for completing semesters with a GPA of 3.9 or higher Fall 2022 - Fall 2025